卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-SG-0201 30-B003701S-S0201

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

水布 专业 第 02 卷 第 01 册
Hydraulic SPECIALITY VOLUME BOOK

卷册名称 站区生活给水管道布置图
Volume name Layout Drawing Of Outdoor Potable Water Piping

批　准 APPROVED BY	
审　核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	罗
图 纸 4 张 DRAWING NOS.	打印件 0 本 PRINTED COPIES NOS.

序号 No.	图　号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
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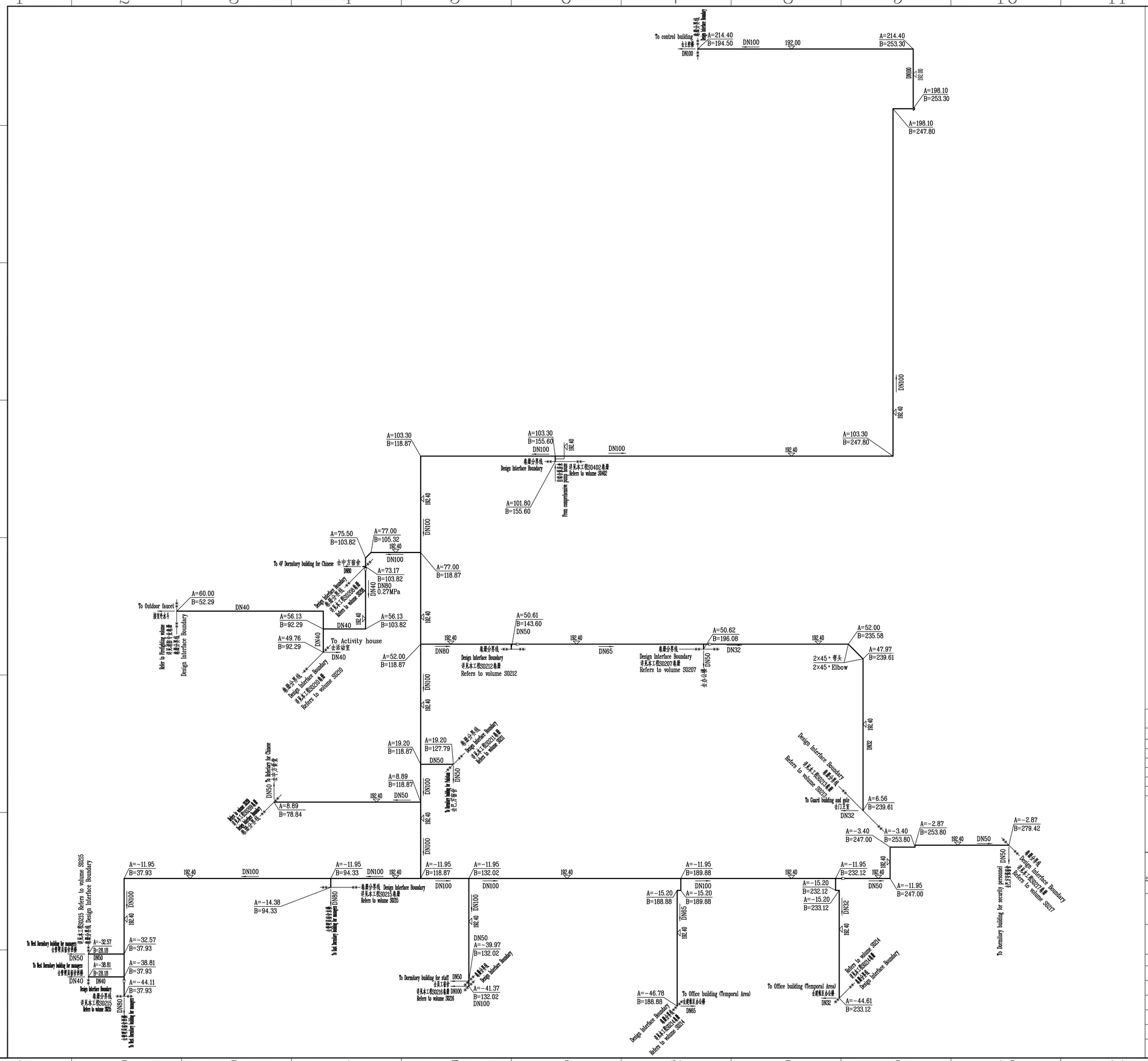
注:



说明：

1、本图卷册标高采用EGM2008高程系统，单位以m计。站区场地平均标高为193.40m，站内道路及换流变广场标高为193.60m。

2、本卷册坐标系统采用站区座标系统，单位以m计；站区座标系统与巴基斯坦UTM-43N平面坐标系的换算公式详见土建相关卷册。

3、站区生活水管采用衬塑镀锌钢管，丝口连接，工作压力为0.40MPa，试验压力为0.90MPa，具体试验要求详见《给水排水管道工程施工及验收规范》(GB50268-2008)。

4、生活水水源从站区给水处理装置区域出水管接出。本工程全站设综合泵房一座，内设50m³水箱一座，生活变频调速供水设备一套。生活变频调速供水设备包括生活水泵两台(一用一备)，参数为Q=25m³/h，H=0.40MPa，N=5.5kW；气压膨胀罐1台，有效容积200L。

5、其他说明详见本卷册02图。

Notes:

1. EGM2008 elevation is adopted in this volume,unit of elevation is m.Average elevation of outdoor ground is 193.40m,and that of road and converter transformer square is 193.60m.
2. The station coordinate(AB) is adopted in this volume,unit of coordinate is m; station coordinate and UTM-43N system conversion formula refers to Civil Engineering volume.
3. Potable water sources from outlet pipe of water treatment plant.A comprehensive pump house shall be set in station area, which contains 50m³ water tank and frequency conversion water supply equipment. Parameters of each potable water pump(totally two,one in use one standby): Q=25m³/h, H=0.40MPa, N=5.5kW; one pneumatic diaphragm tank, effective capacity is 200L.
4. Potable water sources from outlet pipe of water treatment plant.A comprehensive pump house shall be set in station area, which contains 50m³ water tank and frequency conversion water supply equipment. Parameters of each potable waterpump(totally two,one in use one standby): Q=25m³/h, H=0.40MPa, N=5.5kW; one pneumatic diaphragm tank, effective capacity is 200L.
5. Other descriptions refer to drawing S0201-02.

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

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	姓名 Name	日期 Date
设计 Designed	甲	
校核 Checked		
审核 Reviewed		
批准 Approved		

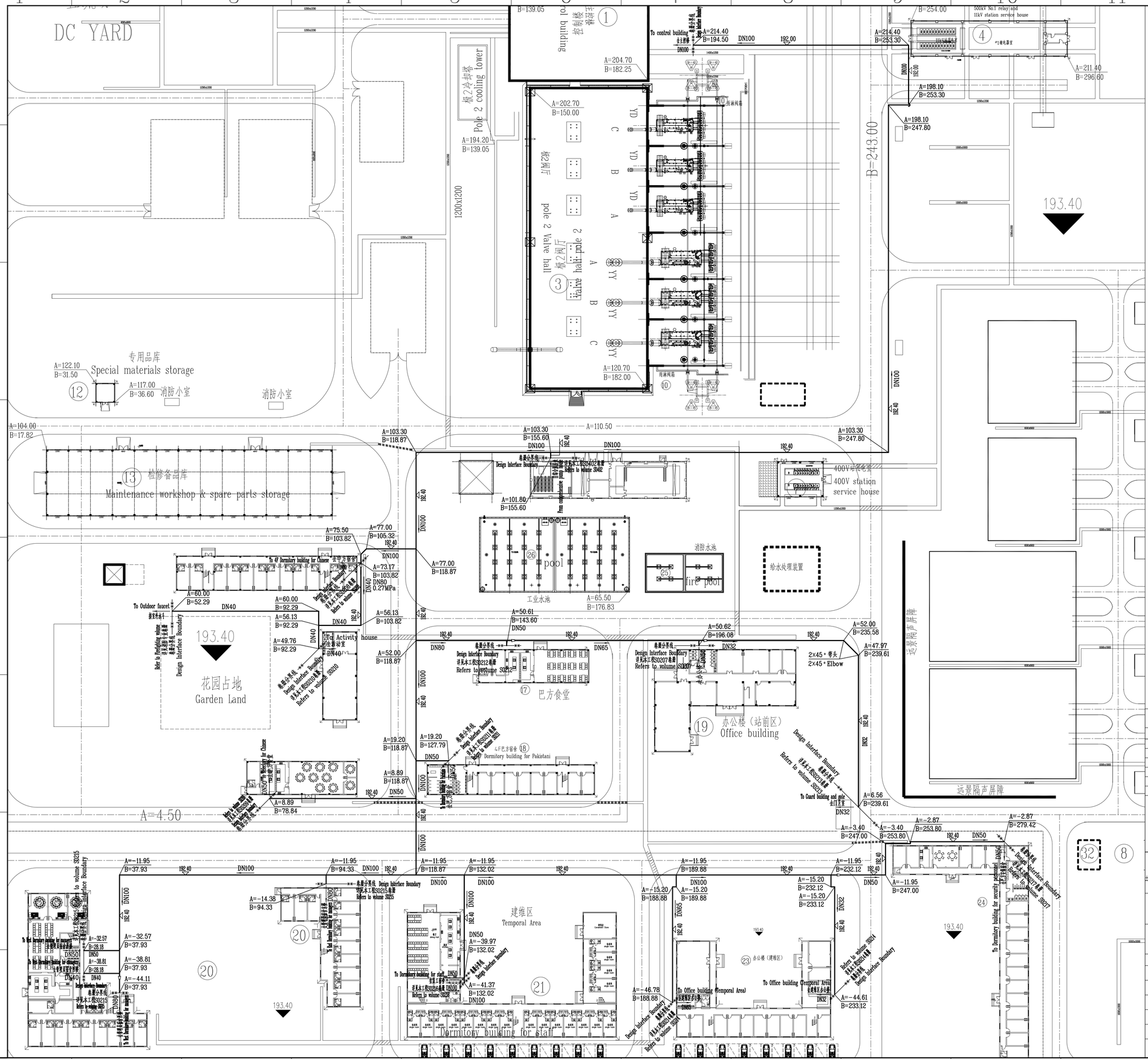
站区生活水系统图
Potable Water Piping System Drawing

比例 Scale: 1:500

图号 DWG NO.: 30-B003701S-S0201-001

版本 REV.: A

594X841



说明：

1、本图格局标高采用EGM2008高程系统，单位以m计。站区场地平均标高为193.40m，站内道路及换流充广场标高为193.60m。
2、本格局座标系统采用站区座标系统，单位以m计；站区座标系统与巴基斯坦UTM-43N平面坐标系的换算公式详见土建相关格局。
3、站内道路为郊区型道路。本图内道路、建筑物、构筑物相对位置仅为示意，以土建相关格局为准。
4、生活水管道防腐等施工说明、管沟交叉典型图详见本工程B003701S-S0403-01图《站区压力直埋管道施工说明》。
5、本期图中管道均采用直埋敷设，所有管道底敷设在150mm厚中粗砂垫层上，安装完毕回填前进行严密性水压试验，水压试验合格后，再覆土回填。
6、站区生活水管采用衬塑镀锌钢管，丝口连接，工作压力为0.40MPa，试验压力为0.90MPa，具体试验要求详见《给水排水管道工程施工及验收规范》(GB50268-2008)。
7、管道施工、安装、回填、试压等按《给水排水管道工程施工及验收规范》GB50268-2008规定执行。
8、生活水水源从站区给水处理装置区域出水管接出。本工程全站设综合泵房一座，内设50m³水箱一座，生活变频调速供水设备一套。生活变频调速供水设备包括生活水泵两台（一用一备），参数为Q=25m³/h，H=0.40MPa，N=5.5kW；气压膨胀罐1台，有效容积200L。
9、室外水龙头的具体位置现场可调整。

Notes:

1. EGM2008 elevation is adopted in this volume,unit of elevation is m.Average elevation of outdoor ground is 193.40m,and that of road and converter transformer square is 193.60m.
2. The station coordinate(AB) is adopted in this volume,unit of coordinate is m; station coordinate(AB) and UTM-43N system conversion formula refers to Civil Engineering volume.
3. Station road adopts suburban road type.Relative location of roads, buildings and structures in the drawing is only for schematic drawing,details refer to Civil Engineering volume.
4. Construction description for pipe corrosion protection and pipeline crossing typical figure refer to figure B003701S-S0403-01《Construction description for directly buried pressure pipeline》.
5. Pipeline in this volume adopts directly buried installation,all piping should be layed on the surface of 150 mm thick coarse sand cushion, hydrostatic test shall be completed before installation,soil backfill shall be done afer water pressure is qualified.
6. Potable water pipeline uses galvanized steel pipe,with threaded connections.Work pressure is 0.40 MPa, test pressure is 0.90 MPa, the specific test requirements refer to《Code for construction and acceptance of water and sewerage pipeline works》(GB50268-2008).
7. Pipeline construction, installation, backfill and pressure test etc shall comply with《Code for construction and acceptance of water and sewerage pipeline works》(GB50268-2008).
8. Potable water sources from outlet pipe of water treatment plant.A comprehensive pump house shall be set in station area, which contains 50m³ water tank and frequency conversion water supply equipment. Parameters of each potable waterpump(totally two,one in use one standby): Q=25m³/h, H=0.40MPa, N=5.5kW; one pneumatic diaphragm tank, effective capacity is 200L.
9. Specific location could be adjusted on site according to the actual situation.

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - S G - 0 2 0 1 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed	中国	
校核 Checked		
审核 Reviewed		
批准 Approved		

站区生活水管道平面布置图
Potable Water Piping Layout Drawing

比例 Scale: 1:500 | 图号 DWG NO.: 30-B003701S-S0201-002 | 版本 REV.: A

594X841

编号 No.	名 称 Name	规 格 Parameters	单位 Unit	数量	材料 Material	单重 Single (重量Weight)	总重 Total (重量Weight)	参考图号及备注 Remark
19	Reducing tee 异径三通	DN100×DN80 PN=1.6Mpa	Set 只	2	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
18	45° Elbow 45°弯头	DN32 PN=1.6Mpa	Set 只	6	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
17	90° Elbow 90°弯头	DN32 PN=1.6Mpa	Set 只	4	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
16	45° Elbow 45°弯头	DN40 PN=1.6Mpa	Set 只	4	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
15	90° Elbow 90°弯头	DN40 PN=1.6Mpa	Set 只	4	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
14	45° Elbow 45°弯头	DN50 PN=1.6Mpa	Set 只	10	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
13	90° Elbow 90°弯头	DN50 PN=1.6Mpa	Set 只	4	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
12	45° Elbow 45°弯头	DN65 PN=1.6Mpa	Set 只	6	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
11	90° Elbow 90°弯头	DN65 PN=1.6Mpa	Set 只	4	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
10	45° Elbow 45°弯头	DN80 PN=1.6Mpa	Set 只	4	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
9	90° Elbow 90°弯头	DN80 PN=1.6Mpa	Set 只	2	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
8	45° Elbow 45°弯头	DN100 PN=1.6Mpa	Set 只	34	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
7	90° Elbow 90°弯头	DN100 PN=1.6Mpa	Set 只	10	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
6	Steel Pipe lined with plastics 衬塑镀锌钢管	D38x2.5(DN32),PN=1.6Mpa	m 米	130	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
5	Steel Pipe lined with plastics 衬塑镀锌钢管	D45x2.5(DN40),PN=1.6Mpa	m 米	150	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
4	Steel Pipe lined with plastics 衬塑镀锌钢管	D57x3(DN50),PN=1.6Mpa	m 米	150	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
3	Steel Pipe lined with plastics 衬塑镀锌钢管	D76x3.5(DN65),PN=1.6Mpa	m 米	40	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
2	Steel Pipe lined with plastics 衬塑镀锌钢管	D89x4(DN80),PN=1.6Mpa	m 米	40	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
1	Steel Pipe lined with plastics 衬塑镀锌钢管	D108x4(DN100),PN=1.6Mpa	m 米	720	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics

编号 No.	名 称 Name	规 格 Parameters	单位 Unit	数量	材料 Material	单重 Single (重量Weight)	总重 Total (重量Weight)	参考图号及备注 Remark
28	Reducer 大小头	DN100×DN65 PN=1.6Mpa	Set 只	1	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
27	Reducer 大小头	DN100×DN80 PN=1.6Mpa	Set 只	1	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
26	Straight tee 等径三通	DN100×DN100 PN=1.6Mpa	Set 只	4	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
25	Reducing tee 异径三通	DN65×DN50 PN=1.6Mpa	Set 只	1	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
24	Reducing tee 异径三通	DN80×DN50 PN=1.6Mpa	Set 只	1	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
23	Reducing tee 异径三通	DN65×DN32 PN=1.6Mpa	Set 只	1	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
22	Reducing tee 异径三通	DN100×DN40 PN=1.6Mpa	Set 只	1	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
21	Reducing tee 异径三通	DN100×DN50 PN=1.6Mpa	Set 只	3	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
20	Reducing tee 异径三通	DN100×DN65 PN=1.6Mpa	Set 只	1	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
编号 No.	名 称 Name	规 格 Parameters	单位 Unit	数量	材料 Material	单重 Single (重量Weight)	总重 Total (重量Weight)	参考图号及备注 Remark

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

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	姓名 Name	日期 Date
设计 Designed	甲	
校核 Checked		
审核 Reviewed		
批准 Approved		
比例 Scale		图号 DWG NO.

站区生活水管道材料表（一）

Bill Of Material For Outdoor Potable Water Piping(1st)

30-B003701C-S0201-003

版本 REV. A

297X420

编号 No.	名 称 Name	规 格 Parameters	单位 Unit	数量	材料 Material	单重 Single	总重 Total	参考图号及备注 Remark
						重量Weight		
31	Reducer 大小头	DN65×DN32 PN=1.6Mpa	Set 只	1	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
30	Reducer 大小头	DN80×DN65 PN=1.6Mpa	Set 只	1	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics
29	Reducer 大小头	DN65×DN50 PN=1.6Mpa	Set 只	1	Q235a衬塑	–	–	DL/T 935-2005 钢塑复合管和管件 Pipes and fittings of steel lined with plastics

编号 No.	名 称 Name	规 格 Parameters	单位 Unit	数量	材料 Material	单重 Single	总重 Total	参考图号及备注 Remark
						重量Weight		

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

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	姓名 Name	日期 Date
设计 Designed	甲	
校核 Checked		
审核 Reviewed		
批准 Approved		
比例 Scale		图号 DWG NO.

站区生活水管道材料表（二）

Bill Of Material For Outdoor Potable Water Piping(2nd)

30-B003701C-S0201-004

版本 REV. A

297X420